

Fetch

Analytics Portfolio

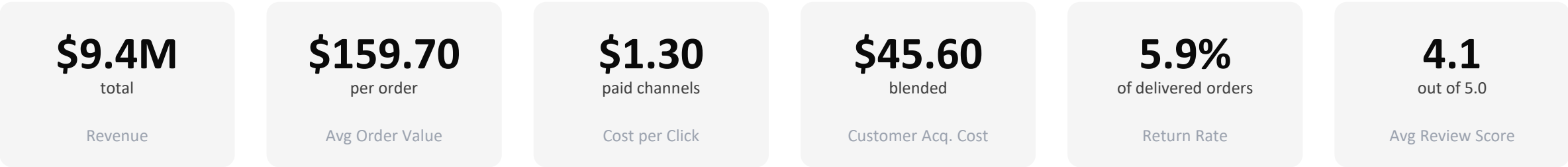
From data collection to customer intelligence

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Business at a Glance

Fetch is a multi-seller e-commerce marketplace connecting thousands of independent sellers to consumers across the US. The business has scaled rapidly but lacks a unified analytics function. This project builds the measurement layer from scratch: warehouse, dashboards, segmentation and predictive models.



Revenue grew from \$46K/month (Oct 2016) to \$985K/month steady state (Aug 2018), with a \$1.15M Black Friday peak

Top 20 product categories account for 85% of revenue; no single category exceeds 9%

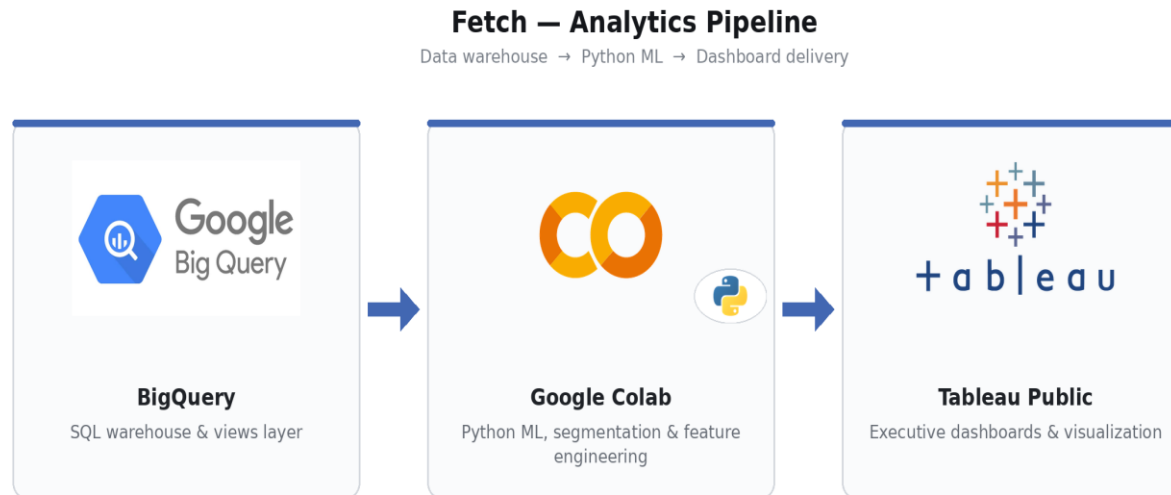
SP state concentration risk: 64.6% of all revenue from one state, top 3 states = 81.5%

97% of customers buy once. Repeat purchase rate is 3% vs industry benchmark of 25-40%

6.77% late delivery rate, but late orders are 6.7x more likely to produce a bad review

Stockout revenue at risk estimated at \$4.5M (29% of realised revenue)

Tech Stack and Data Pipeline



Warehouse

BigQuery hosts 8 dimension and fact tables, 15 analytical views, and the ad-spend join.

Compute

Google Colab runs Python for ad-spend analysis, K-Means segmentation and predictive modelling.

Visualisation

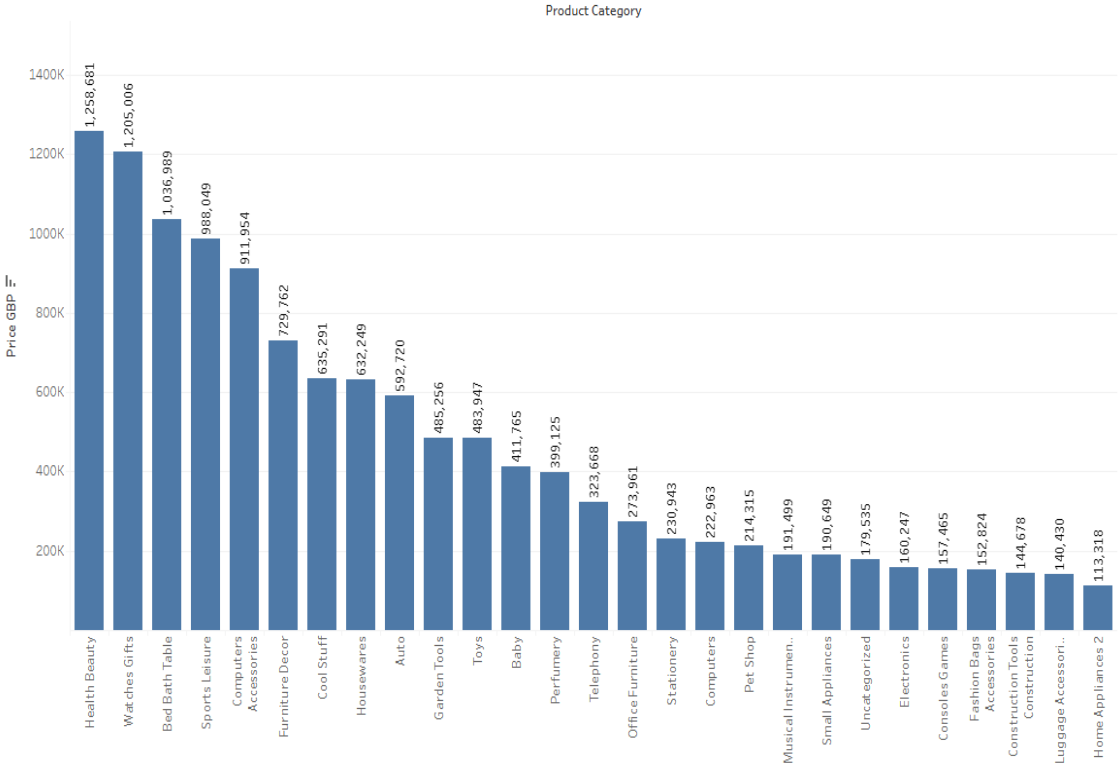
Tableau Public serves 4 department dashboards with live extracts.

Data quality

Three documented issues found and resolved during exploration (review duplicates, week alignment, delivery timestamp batching).

Revenue Performance

Revenue by Category +90,000



Key findings

Revenue grew from \$46K/month (Oct 2016) to \$985K/month steady state, with a \$1.15M Black Friday peak in Nov 2017

Top 20 categories account for 85% of revenue. Health Beauty leads at 9%, with no single-category dependency

AOV flat at \$155-170 across all 7 channels, so channel strategy is a volume and cost decision, not a quality one

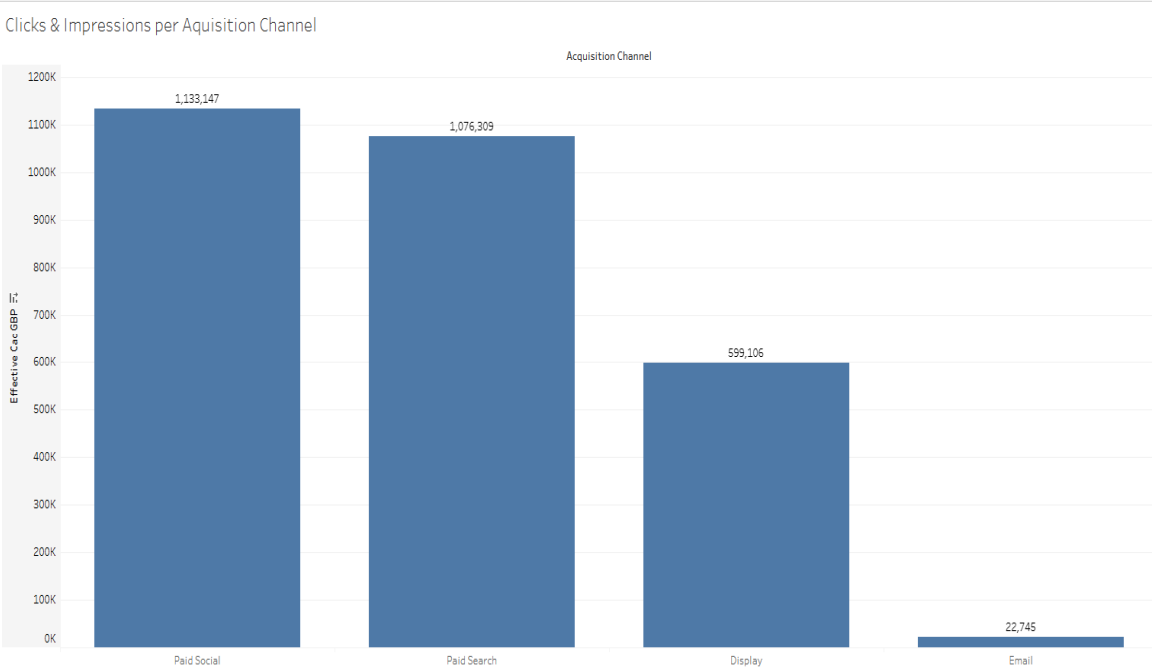
SP state concentration risk: 64.6% of revenue from one state. Top 3 states = 81.5%

Dec 2016 anomaly: 1 order only, likely a marketing pause or platform outage

Top seller accounts for 1.6% of revenue; top 20 = 22%. No seller concentration risk across 3,000+ sellers

Seller strategy split: volume plays (~\$80 AOV, 1,800+ orders) vs high-ticket (~\$660 AOV, few hundred orders)

Marketing Efficiency



Channel efficiency

Email CAC: \$2.20 (lowest cost channel, but small volume)

Paid Search CAC: \$43.83 (primary volume driver)

Paid Social CAC: \$57.85 (mid-tier efficiency)

Display CAC: \$76.47 (highest cost, weakest return)

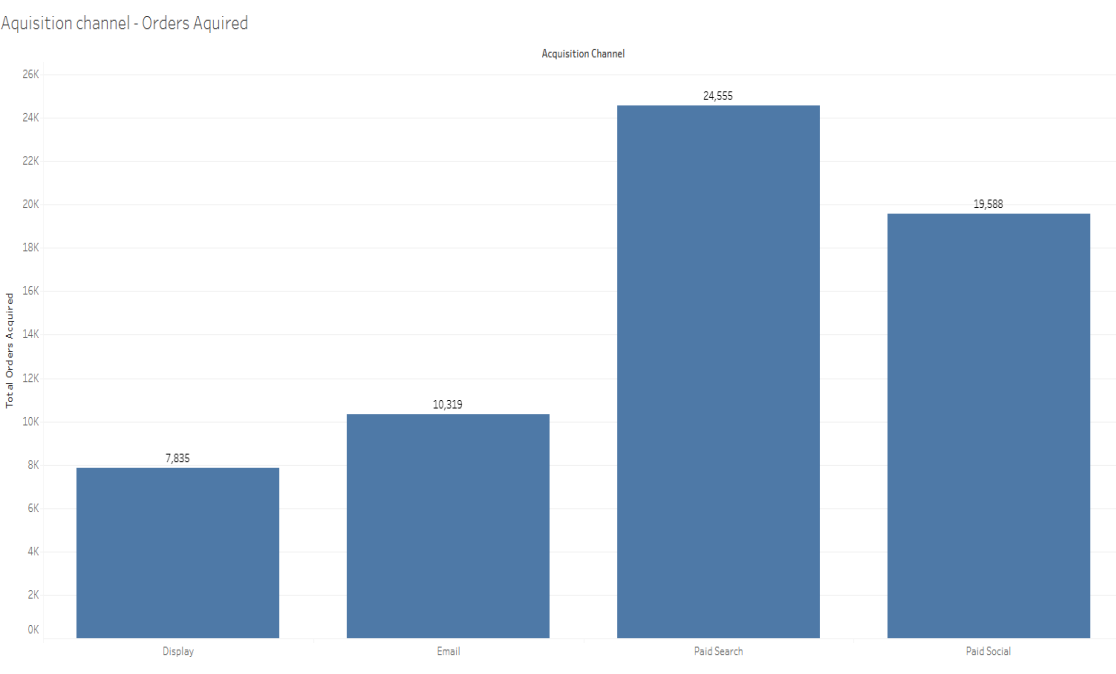
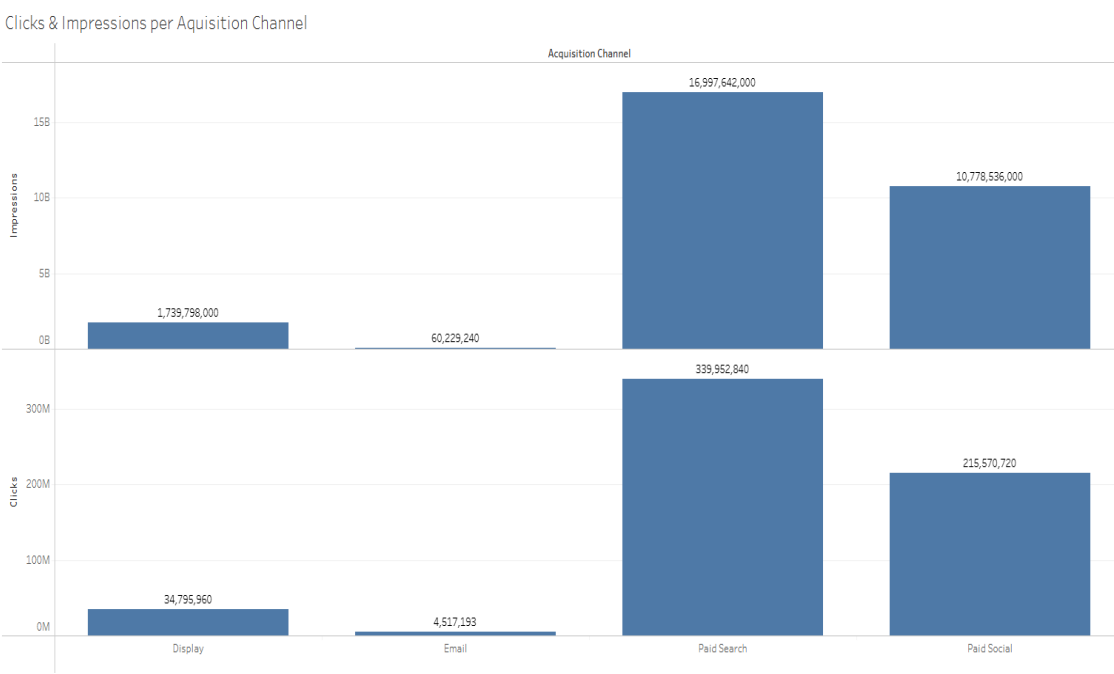
CTR (2.0%) and conversion rate (2.5%) are identical across all paid channels: CAC differences are driven entirely by CPM, not funnel quality

The strategic lever is CPM negotiation or shifting budget toward Email, which has 20x better efficiency than Display

Diagnostic funnel

Channel	CTR %	Conv %	CPM	CAC
Email	7.5	33.3	\$55	\$2.20
Paid Search	2.0	2.5	\$22	\$43.83
Paid Social	2.0	2.5	\$29	\$57.85
Display	2.0	2.5	\$38	\$76.47

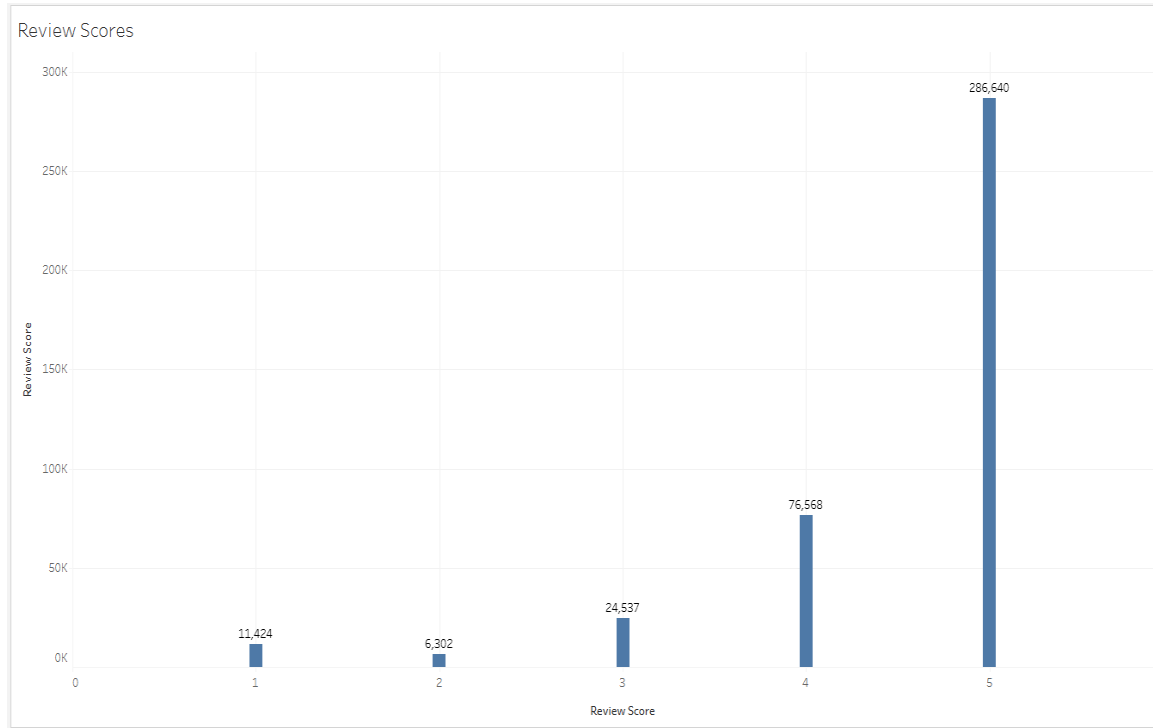
Channel Attribution



Organic overtook Paid Search as the top channel from April 2018, a sign of brand maturity.

Direct traffic grew 40x over the period, acting as a proxy for brand recognition. CTR (2.0%) and conversion rate (2.5%) are identical across paid channels, which means CAC differences are driven entirely by CPM, not funnel quality. The strategic lever is CPM negotiation or shifting budget toward lower-cost channels like Email (\$2.20 CAC).

Operations and Delivery



Delivery and CSAT

Median delivery: 10 days. Mean 12.5 (right-skewed). P90 = 23 days, meaning 10% of customers wait 3+ weeks

On-time rate: 93.2%. Within 2 points of Amazon Prime benchmarks, without centralised logistics

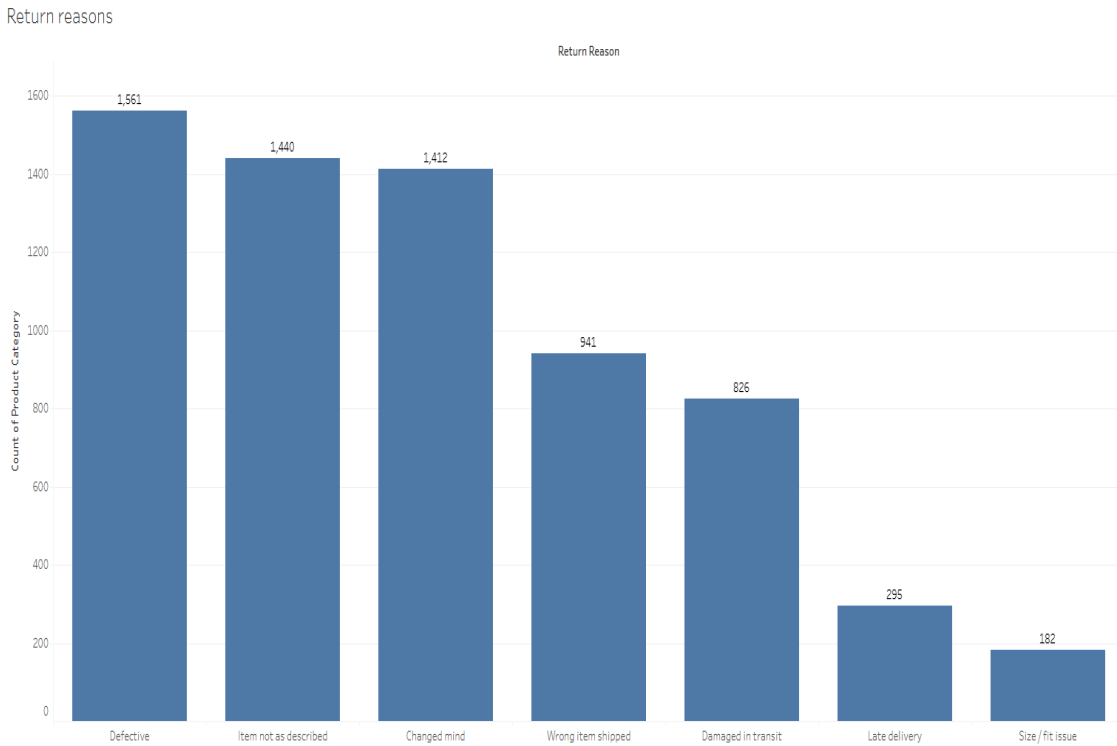
Late orders score 2.27 avg review (vs 4.29 on-time). 62.4% of late orders produce a bad review, a 6.7x multiplier

Late delivery drives 28% of all bad reviews from only 6.77% of orders. Cutting late rate by 50% eliminates 14% of all negative reviews

NPS proxy: +34.85. 57.8% five-star, 11.5% one-star. The 1-star bucket is 3.6x the 2-star bucket: customers polarise rather than settle in the middle

Data quality note: 14 of top 20 longest deliveries timestamped in a single 4-hour batch window (reconciliation artefact, not real wait times)

Returns and Seller Quality

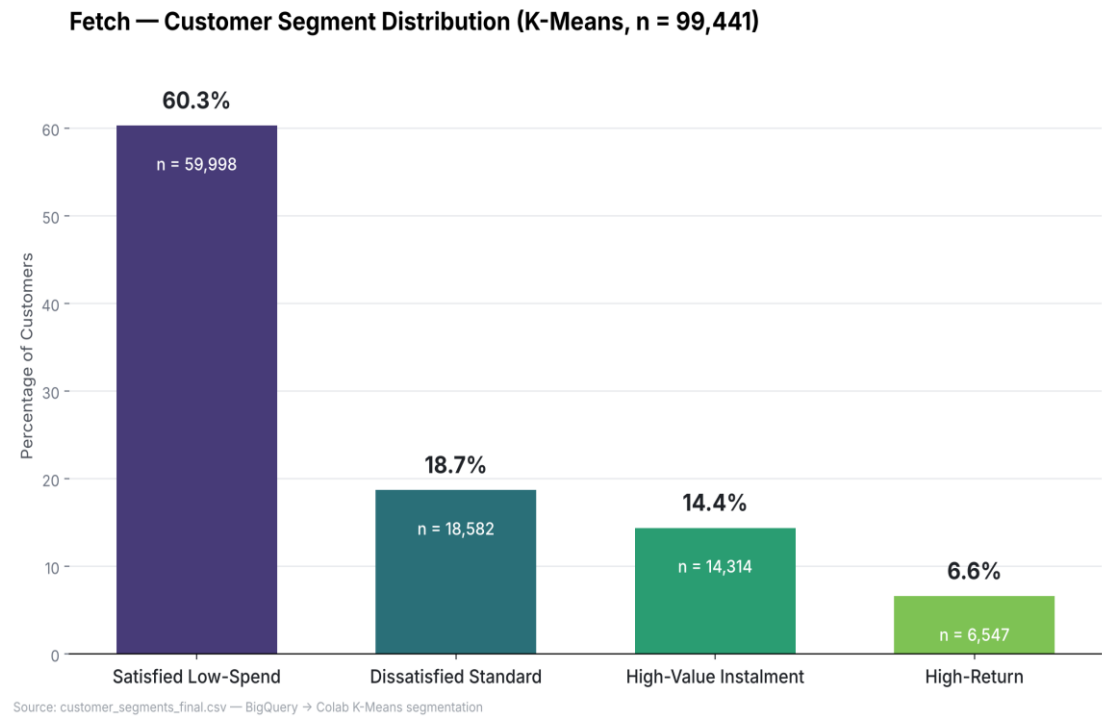


59% of returns are seller-preventable

● Defective product	23.5%
● Item not as described	21.6%
● Changed mind	21.2%
● Wrong item shipped	14.1%
● Damaged in transit	12.4%
● Late delivery	4.4%
● Size/fit issue	2.7%

*Behavioural contradiction: late delivery drives 28% of bad reviews but only 4.4% of returns.
Customers keep late items but punish the seller in ratings.*

Customer Intelligence: K-Means Segmentation



97% of customers buy once

Repeat purchase rate is 3% vs industry benchmark 25-40%. Growth is entirely acquisition-driven. Moving repeat rate from 3% to 8% would double the revenue base without new acquisition spend.



- Satisfied Low-Spend**

60.3% Low frequency, low monetary, high satisfaction. Largest group.
- Dissatisfied Standard**

18.7% Average spend, poor reviews. Churn risk.
- High-Value Instalment**

14.4% Higher basket size, uses instalment payments.
- High-Return**

6.6% Elevated return rates. Operational cost driver.

Strategic Recommendations

01

Fix Retention

97% one-and-done rate is the single largest strategic gap. A post-purchase re-engagement flow (email sequences, personalised recommendations via segment data) targeting the Satisfied Low-Spend cluster (60% of customers) has the highest expected return. Moving repeat rate from 3% to 8% doubles the active revenue base.

3% repeat rate

Target: 8%

02

Reduce Late Deliveries

6.77% of orders arrive late, but those orders are responsible for 28% of all bad reviews. The CSAT multiplier is 6.7x. A 50% reduction in late deliveries eliminates 14% of negative reviews and materially improves the NPS proxy (+34.85 today). Priority: identify the worst-performing seller-state-carrier combinations.

6.77% late rate

Target: 3.5%

03

Address Seller Quality

59% of returns are seller-preventable (defective, misdescribed, wrong item). Stockout events represent \$4.5M in revenue at risk (~29% of realised revenue). Introduce a seller quality scorecard combining return reason mix, review scores and stockout frequency. Gate marketplace visibility on the score.

59% seller-fault returns

Target: 30%

Summary and Next Steps

What we built

8 warehouse tables and 15 analytical views in BigQuery

15-query descriptive narrative covering sales, marketing, operations and customer behaviour

4 Tableau dashboards (Revenue, Marketing, Operations, KPI Overview)

Ad-spend efficiency analysis across 4 paid channels with diagnostic funnel

K-Means customer segmentation (4 clusters on RFM features)

3 data quality issues documented and resolved

Next steps

Predictive model: churn probability scoring using segment features and order history

CLV estimation per segment to prioritise retention spend

Seller quality scorecard integrating return reasons, reviews and stockout data

Delivery carrier performance analysis by state to identify worst-case routes

Geo-translation layer (US state mapping) for portfolio presentation

Fetch is building its analytics capability from the ground up. This is the foundation.